

PREDICTION OF ONLINE ADDICTION OF STUDENTS DURING LOCKDOWN USING MACHINE LEARNING METHODS

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November, 2021.

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The undersigned hereby certify that they have read and recommend to the Computer Science and Engineering Department for acceptance a project entitled “**Prediction of online addiction of students during lockdown**” in partial fulfillment of the requirements for the degree of **B.Sc. Engg.(CSE)**.

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Title: **prediction of online addiction of students during lockdown**

Department: **Computer Science and Engineering**

Degree: **Bachelor of Science in Computer Science and Engineering**

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ACKNOWLEDGEMENT

First I express my heartiest thanks and gratefulness to almighty God for His divine blessing makes me possible to complete the final year project successfully.

I really grateful and wish my profound my indebtedness to Md. Mehedi Rahman Rana, Lecturer Department of CSE, Northern University of Business & Technology, Khulna. His infinite patience, scholarly guidance, continual encouragement, constant and energetic supervision, constructive criticism, valuable advice, reading many inferior draft and correcting them at all stage have made it possible to complete this project documentation. Without this valuable support and guidance, this internship could not elevate up to this level of development from my point of view.

I would like to thank our entire course mate in Northern University of Business & technology, Khulna who took part in this discuss while completing the course work.

Finally, I must acknowledge with due respect the constant support and patients of my parents.

Abstract

Online addiction of students is a very common problem now-a-days. In this lockdown, this problems increases a lot. My thesis purpose is that, prediction of online addiction of students during lockdown. So,i was collected 500 students real datasets by online survey.I was created a online google form and set up the questionnaires on that google form and then send that google form link to the students by using online social platforms.when five hundred responses fulfill on the google form,then I was downloaded the csv file from there .Then i was deleted the irrelevant things from the dataset and customize the datasets properly.I was used level encoding for converting categorical data to the numeric data.I was used three machine learning algorithms like: random forest, dicision tree, logistic regression algorithms on my code.In random forest algorithm I was get 96% accuracy,in logistic regression algorithm I was get 94% accuracy,in dicision tree algorithm i was get 96% accuracy.

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CHAPTER I

Introduction

1.1 Overview

- Online addiction of students is a very common problem now-a-days. In this lockdown, this problems increases a lot. My thesis purpose is that, prediction of online addiction of students during lockdown. So, i was collected 500 students real datasets by online survey. I was created a online google form and set up the questionnaires on that google form and then send that google form link to the students by using online social platforms. when five hundred responses fulfill on the google form, then i was downloaded the csv file from there. Then i was deleted the irrelevant things from the dataset and customize the datasets properly.
- I was used level encoding for converting categorical data to the numeric data. I was used three machine learning algorithms like: random forest, decision tree, logistic regression algorithms on my code. In random forest algorithm i was get 96% accuracy, In logistic regression algorithm i was get 94% accuracy, In decision tree algorithm i was get 96% accuracy. I was showed the graph of how much time does student spend online without any kind of productive work(in hour) on my code.

1.2 Motivation

- Prediction of online addiction of students during lockdown provides the necessary statistics of how many time does students spend online without any productive work(in hour).
- In lockdown, students were getting so much addicted at online & they were spending there valuable time at online without doing any kind of productive work.
- My motivation is that, if i predict the statistics of how many time does student spend online without any productive work(in hour) in lockdown it will become necessary work for the future young generation.

1.3 Objective of the study

- The main objective of my study is that, to predict how much time does student spend online without any productive work(in hour) & to compare my proposed systems features.
- The count of students spend online without any productive work showed on the below graph. In the graph, represented that 2-4hours & more than 6hours graph lines are maximum counted & the 1-2hours, none, 4-6hours graph lines represents the minimum counts.
- Compare between my proposed system features like spend online without any productive work feature was compared with the age feature.

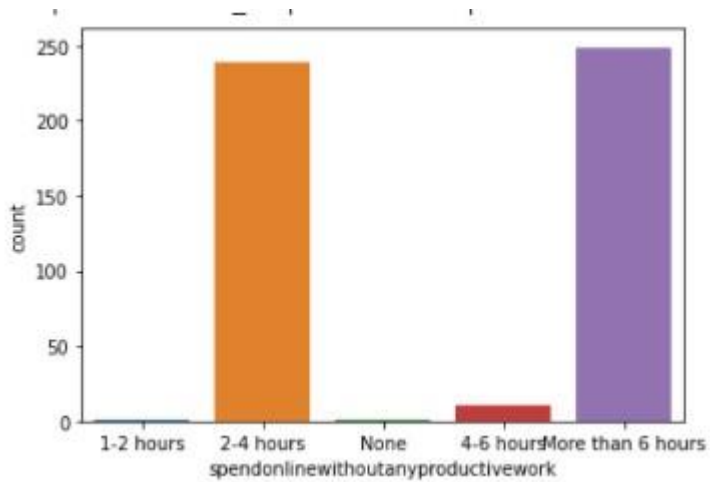


Fig 1 : How much time does student spend online without any productive work(in hour).

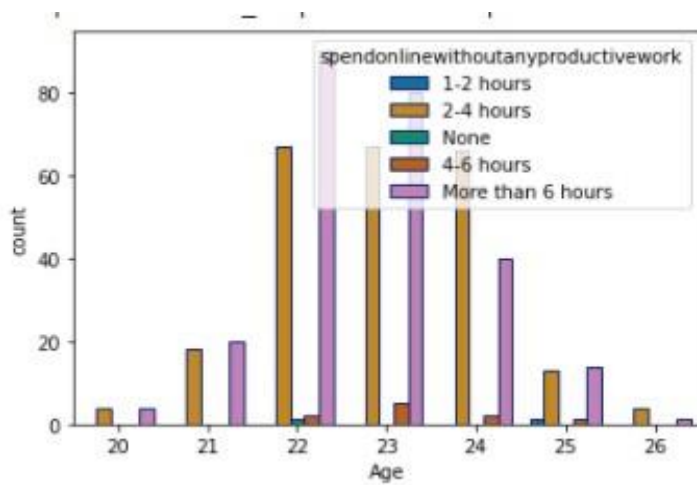


Fig 2 : Compare spend online without any productive work feature with age feature.

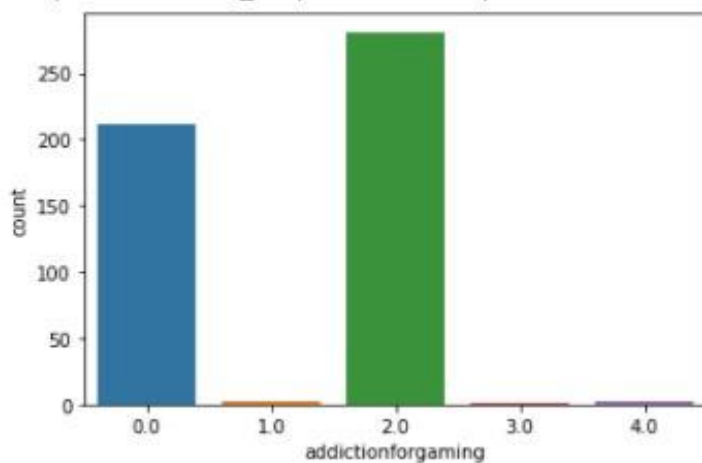


Fig 3 : Compare addiction for gaming feature with spend online without any productive work feature.

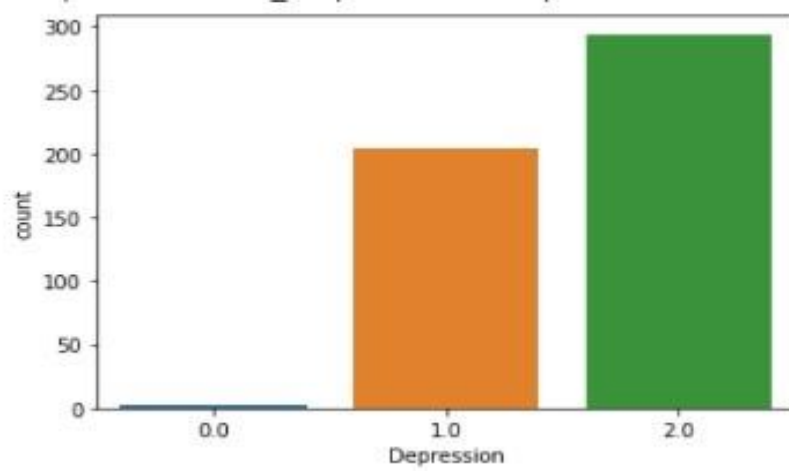


Fig 4 : Compare depression feature with spend online without any productive work feature.

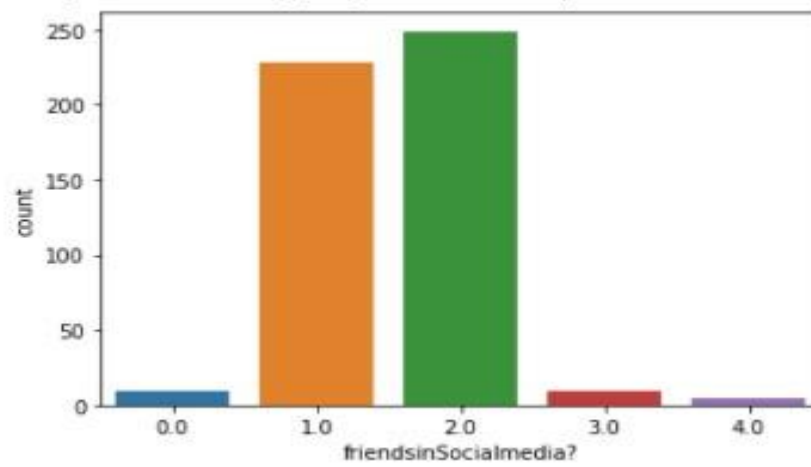


Fig 5 : Compare friends in social media feature with spend online without any productive work feature.

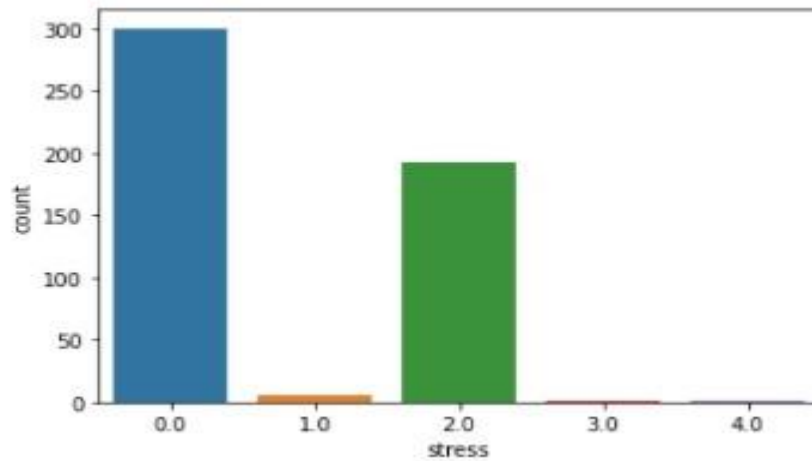


Fig 6 : Compare stress feature with spend online without any productive work feature.

<matplotlib.axes._subplots.AxesSubplot at 0x7f72d982fa10>

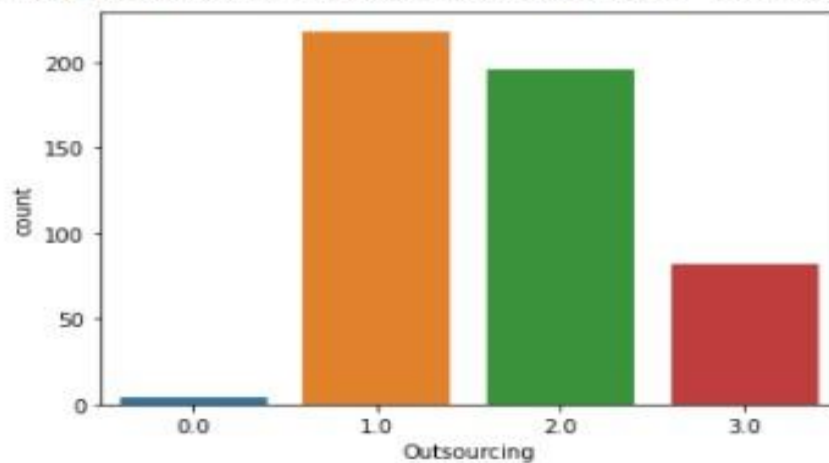


Fig 7 : Compare outsourcing feature with spend online without any productive work feature.

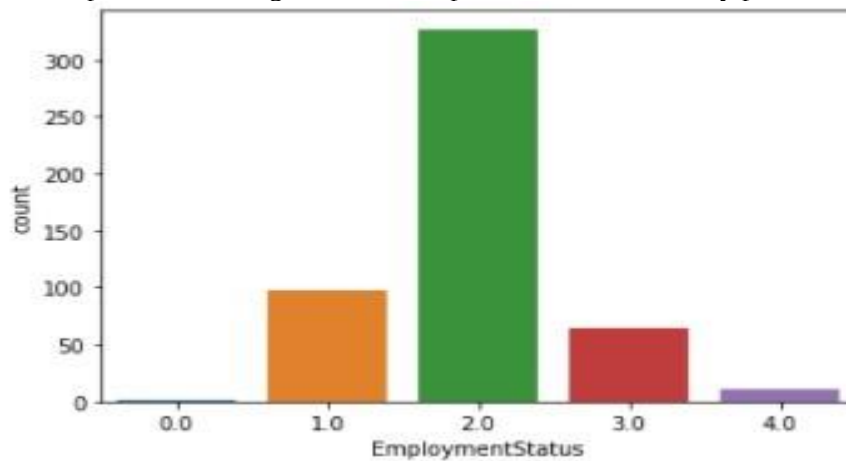


Fig 8 : Compare Employment status feature with spend online without any productive work feature.

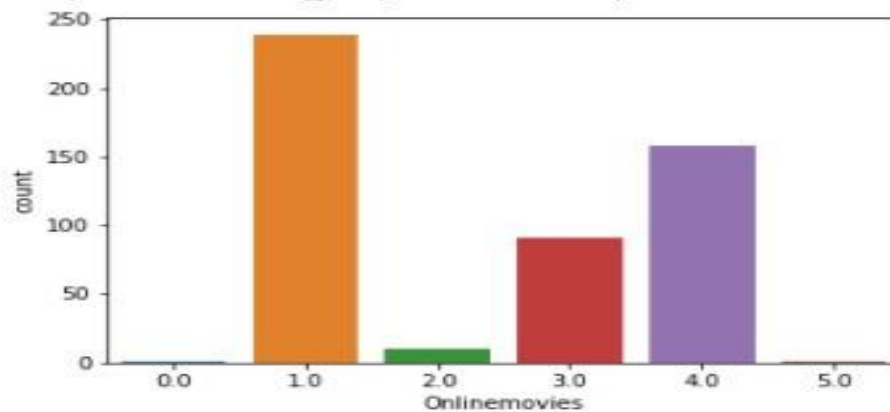


Fig 9 : compare online movies feature with spend online without any productive work feature.

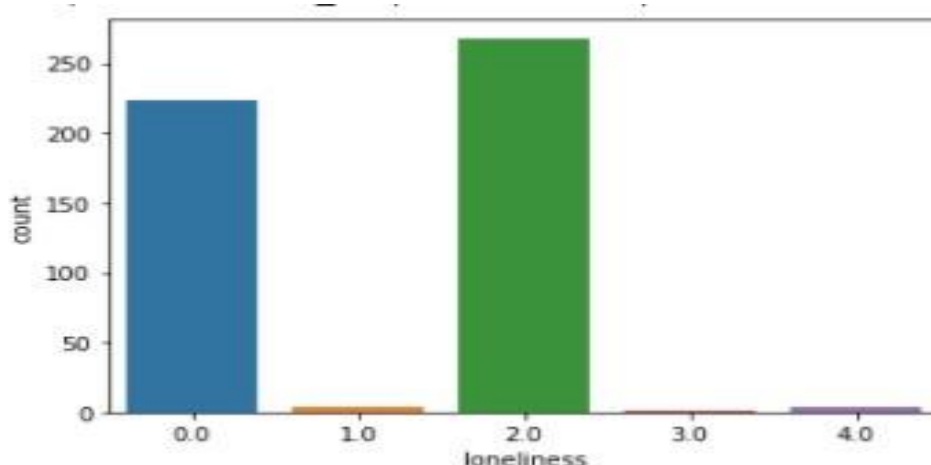


Fig 10 : compare loneliness feature with spend online without any productive work feature.

CHAPTER I I

REVIEW OF LITERATURE

- In existing system work,they compared their datasets features.[1]In my proposed system work,i also compared my datasets features.My target feature was spend online without any productive work.I was compared my other features with my target features.
- In existing system work,they were showed students gaming addiction percentage.[1]In my proposed system i showed that for gaming addiction how much time does student spend online without any productive work.
- In existing system work,they were showed student addiction for social media percentage.[1]In myProposed system, i showed that for social media addiction how much time does student spend online without any productive work.
- In existing system work,they were showed student stress percentage.[2]In my proposed system, i showed that for spend online without any productive work how much stress students suffered.
- In existing system work,they were showed students depression percentage.[3]In my proposed system, i showed that for spend online without any productive work how much depression students suffered.
- In existing system work,they were showed students psychological problems.[4]In my proposed system,i showed that for spend online without any productive work how much psychological problems faced students.
- In existing system work they were showed students educational level percentage[5]In my proposed system,i was showed that spend online without any productive work students education level.
- In existing system work they were showed students loneliness percentage[6]In my proposed system,i was showed spend online without any productive work students loneliness.
- In existing system work they were showed students employment status percentage[7]In my proposed system, i was showed that spend online without any productive work students employment status.

- In existing system work they were showed students outsourcing percentage[8]In my proposed system, i was showed that spend online without any productive work students outsourcing percentage.
- In existing system work they were showed students addiction for online movies percentage[9]In my proposed system,i was showed that spend online without any productive work students addiction for the online movies percentage.
- In existing system work they were showed students addiction for different messaging apps[10]In my proposed system,i was showed that spend online without any productive work students different messaging apps used percentage.
- In existing system, work they were showed students total online addiction percentage.[11]In my proposed system, i was showed that how much time does student spend online without any productive work.
- In existing system work they were showed students friends in social media percentage.[12]In my proposed system, i was showed that spend online without any kind of productive work students friends in the social media percentage.
- In existing system work they showed student anxiety percentage[13]In my proposed system, i was showed spend online without any productive work students anxiety percentage.
- In existing system work they showed that students family apathy problems percentage[14]In my proposed system,i was showed spend without any productive work students family apathy problems percentage.
- In existing system work they showed that students external affairs percentages[15]In my proposed system, ,i was showed spend online without any productive work students external affairs percentages.
- In existing system work they showed students having computer percentages[16]In my proposed system,i was showed spend online without any kind of productive work students having computer percentages.

CHAPTER III

METHODOLOGY

- I was collected 500 students real datasets by online survey.
- I was created a online google form and set up the questionnaires on that google form and then send that google form link to the students by using online social platforms.

- when five hundred responses fulfill on the google form, then I was downloaded the csv file from there .
- I was used level encoding for converting categorical data to the numeric data.

Name	Age	user	gender
KabirEfaz	25	WiFi- Broadband	Male
Razu	22	WiFi- Broadband	Male
Nishat Imran Sanjana	22	WiFi- Broadband	Female
Imran	23	WiFi- Broadband	Male

Table 1 : Categorical data

Name	Age	user	gender
KabirEfaz	25	2.0	1.0
Razu	22	2.0	1.0
Nishat Imran Sanjana	23	2.0	0.0
Imran	22	2.0	1.0

Table 2 : Numeric data

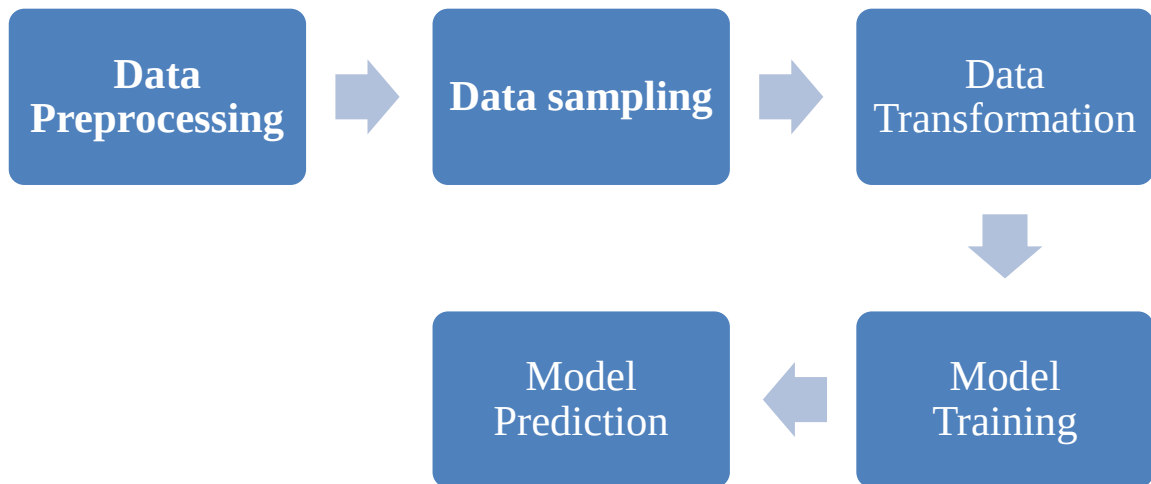


Fig 11 : Proposed Methodology

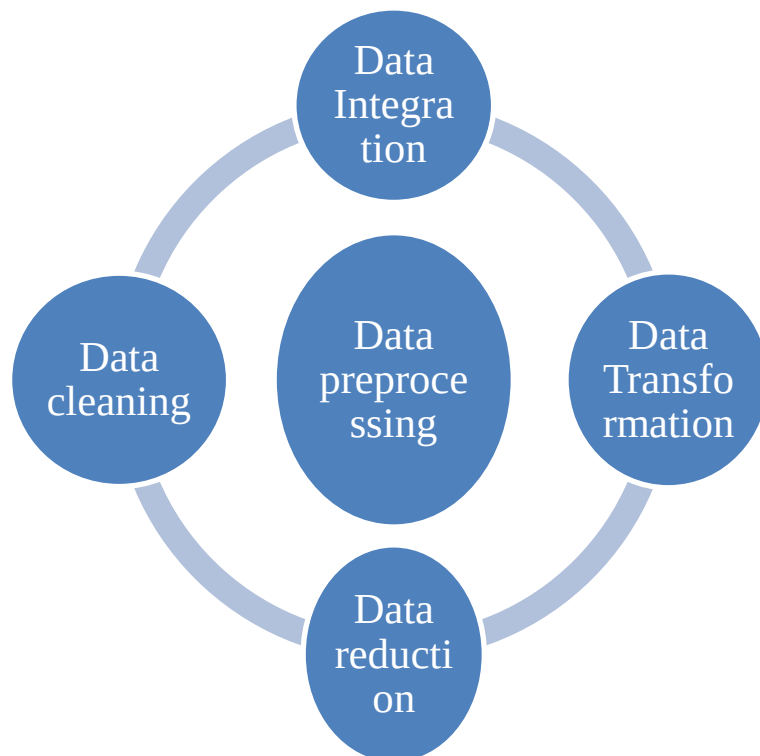


Fig 12 : Data preprocessing steps

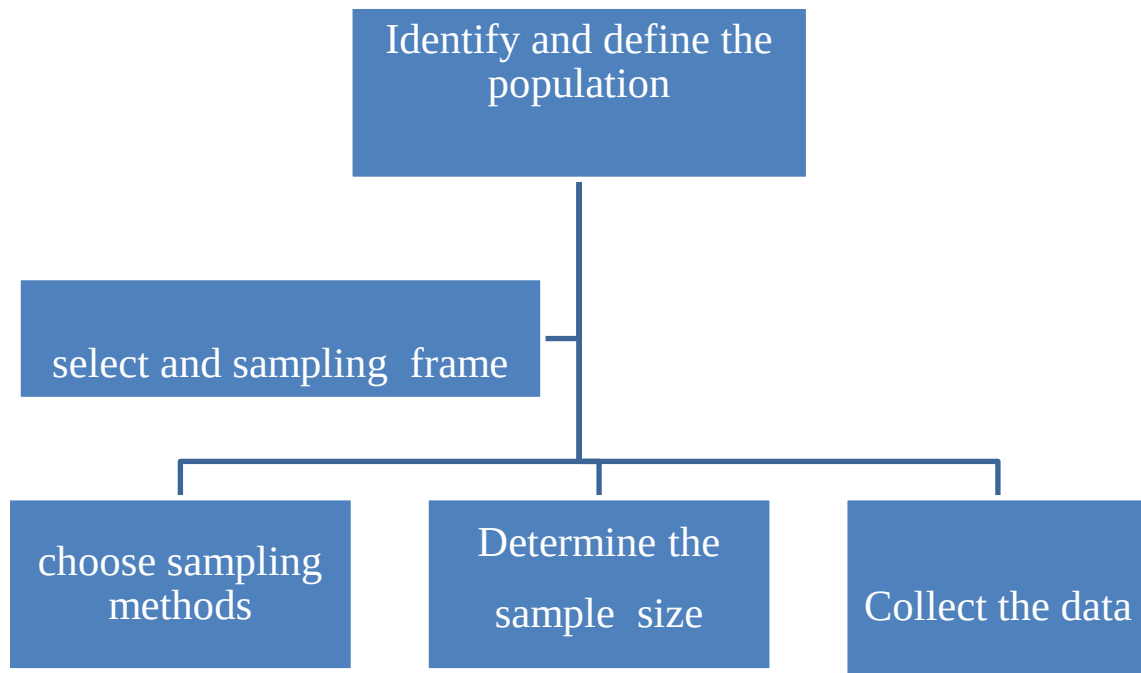


Fig 13 : Data sampling steps

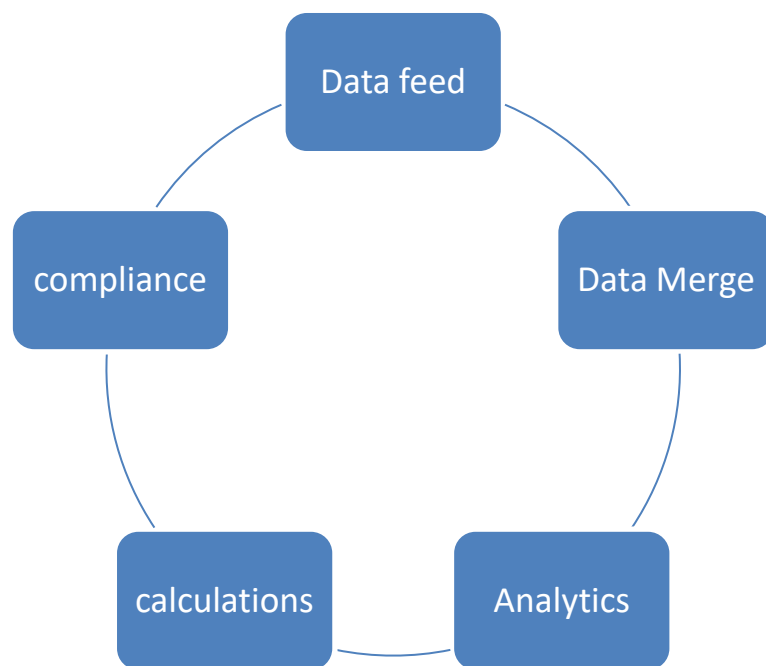


Fig 14 : Data transformation steps

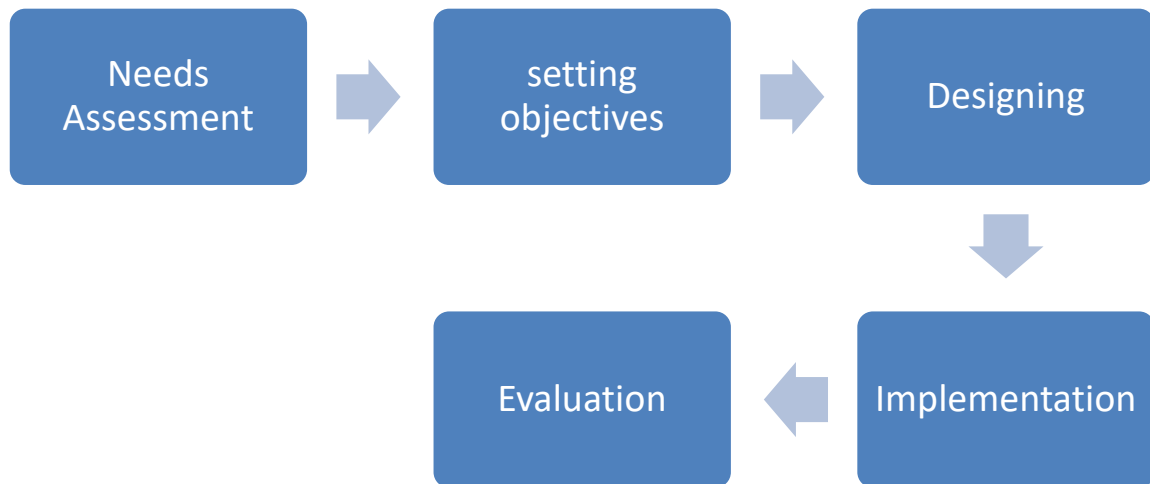


Fig 15 : Model Training steps

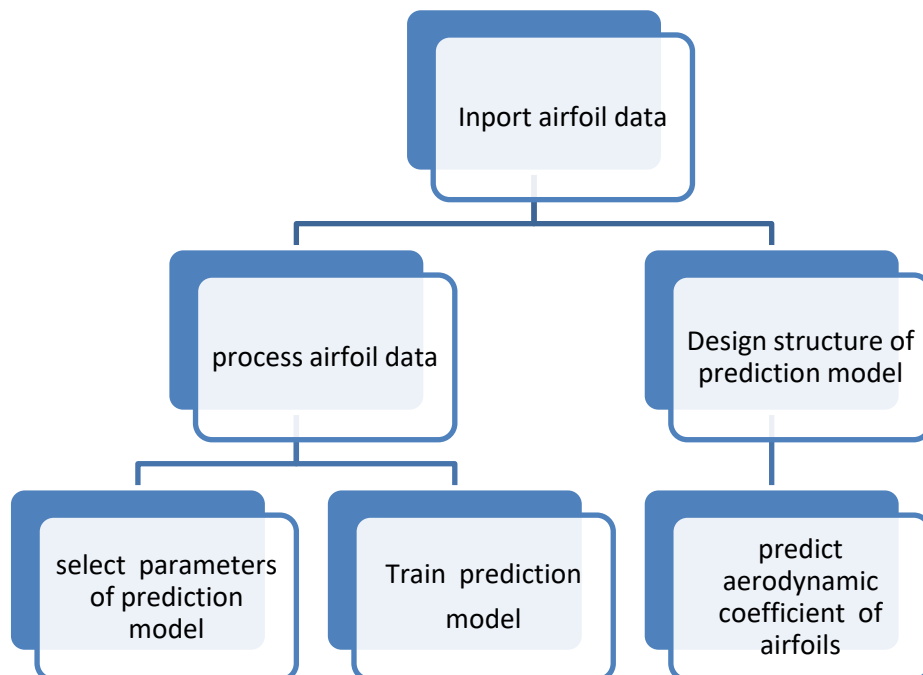


Fig 16 : Model prediction steps

CHAPTER IV

RESULTS AND DISCUSSION

- .I was used three machine learning algorithms like: random forest, dicision tree, logistic regression algorithms on my code.
- In random forest algorithm i was get 96% accuracy.
- In logistic regression algorithm i was get 94% accuracy.

- In decision tree algorithm i was get 96% accuracy.

Algorithm	Accuracy
Logistic Regression	94%
Decision Tree	96%
Random Forest	96%

CONFUSION MATRIX

		Actual	
Predicted		Positive	Negative
	Positive	39	10
	Negative	20	81

Fig 17 : confusion matrix using random forest algorithm

Actual			
Predicted		Positive	Negative
	Positive	40	22
	Negative	16	72

Fig 18 : confusion matrix using decision tree algorithm

Actual			
Predicted		Positive	Negative
	Positive	45	10
	Negative	11	46

Fig 19 : confusion matrix using logistic regression algorithm

CHAPTER V

Conclusion

- During lockdown period most of the students spend online without any kind of Productive work.
- Just because of that most of the students are not properly use their Valuable times .
- Students are getting suffered by lots of various types of diseases just like depression, anxiety, stress, lots of problems which actually damage their mental health condition.
- Family should aware about this problem & give the proper supports & guidance.

Limitation And Future Work

There are some drawbacks in our system.

This prediction of Online addiction of students during lockdown was just for Bangladesh Country , Not for Others Countries.

In future, we predict also other countries this kind of Statistics with much better accuracy .

CHAPTER VI

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